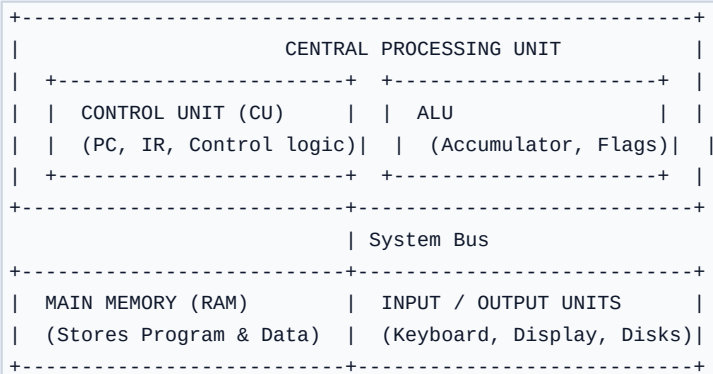


3-Mark Questions & Answers -- MPCA Unit 1: Overview of Computer Architecture & Organization

Q1. Explain the Von Neumann model with a neat diagram.

The Von Neumann Model describes a computer system based on stored-program concepts, where instructions and data reside in a single shared memory.



* Control Unit (CU): Fetches instructions and directs operations.

* AL

Q2. Explain any three addressing modes with examples.

1. Immediate Addressing Mode: The operand is specified directly within the instruction itself.

* *E

Q3. Differentiate between CISC and RISC architecture.

Feature	CISC (Complex Instruction Set)	RISC (Reduced Instruction Set)
Instruction Count	Large set (300+ complex instructions).	Small set of simple, uniform instructions.
Execution Time	Variable (multiple clock cycles per instruction).	Fixed (single clock cycle per instruction).
Memory Access	Operations allowed directly on memory.	Load/Store architecture only (operands in registers).
Examples	Intel x86 (8086, Core i9), AMD Ryzen.	ARM, MIPS, RISC-V.

Q4. Explain the evolution of computers (generations) briefly.

1. 1st Gen (1940-1956): Vacuum tubes; massive physical size; machine language; magnetic drums.

2. 2nd

Q5. What are the different types of instruction formats based on operand count

1. Three-Address Instructions: Specify 2 source operands and 1 destination operand.

* *S

